

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SCENARIO BASED QUESTIONS

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
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Weightage:	40% of CIA	Total Marks:	100
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Code of Conduct

I V BOOPATHY KHANISHK(192524311) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ KHANISHK V B _____

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO4 – AT1

SCENARIO BASED QUESTIONS

HYPOTHESIS TESTING USING P-VALUE AND CONFIDENCE INTERVAL

Case Studies 17 – 20

DSA04 – Fundamentals of Data Science

Presented by

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AUGUST 22, 2026

CERTIFICATE

This is to certify that V. Boopathy Khanishk (192524311), student of the Department of Computer Science and Engineering, has completed the case study report titled “Hypothesis Testing Using p-value and Confidence Interval” for DSA04 – Fundamentals of Data Science. The report covers the supplied case studies on hypothesis testing, p-values, and confidence intervals.

The report presents the statistical reasoning, calculations, interpretations, and practical implications associated with the problems and is prepared for academic evaluation.

Dr. KRISHNARAJ MONY

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DECLARATION

I hereby declare that this case study report titled “Hypothesis Testing Using p-value and Confidence Interval” has been prepared as part of the academic requirements for DSA04 – Fundamentals of Data Science. The report presents systematic solutions to the supplied case studies using the statistical information given in the questions.

The interpretations are intended for academic purposes and emphasize correct use of p-values, significance levels, confidence intervals, and practical decision making.

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1. INTRODUCTION TO HYPOTHESIS TESTING

Hypothesis testing is a statistical method used to evaluate a claim about a population using evidence collected from a sample. In data science, hypothesis testing is useful because decisions are frequently made from incomplete information. Analysts may observe a sample of customers, products, transactions, system responses, or model predictions and then determine whether the observed evidence is strong enough to support a conclusion about the larger population.

A hypothesis test normally begins with two competing statements. The null hypothesis, H_0 , describes a baseline assumption or status quo. The alternative hypothesis, H_1 or H_a , represents the effect, difference, or directional claim being investigated.

The significance level, α , is selected before the evidence is evaluated. Common choices include 0.05 and 0.01. It provides the threshold for deciding whether the observed evidence is sufficiently inconsistent with the null hypothesis.

The p-value is compared with α . If $\text{p-value} \leq \alpha$, the null hypothesis is rejected. If $\text{p-value} > \alpha$, the null hypothesis is not rejected. The second conclusion does not prove that the null hypothesis is true; it means that the available evidence is not sufficiently strong to reject it.

The case studies in this report apply these ideas to software response time, advertising click-through rate, machine-learning accuracy, factory defects, and customer spending.

2. UNDERSTANDING THE P-VALUE

A p-value measures how compatible the observed sample evidence is with the null hypothesis. More precisely, assuming H_0 is true, it is the probability of obtaining a test statistic at least as extreme as the observed statistic in the relevant direction.

A small p-value indicates that the observed result would be relatively unusual if H_0 were true. Therefore, a sufficiently small p-value provides evidence against H_0 .

A p-value is not the probability that H_0 is true. For example, $p = 0.004$ does not mean that H_0 has a 0.4% probability of being true. It means that, under H_0 , results at least as extreme as the observed result have probability 0.004.

$$\mathbf{p\text{-}value \leq \alpha \rightarrow \text{Reject } H_0}$$

$$\mathbf{p\text{-}value > \alpha \rightarrow \text{Fail to reject } H_0}$$

Statistical significance should also be distinguished from practical significance. A statistically significant result may have a small practical effect, while a practically important effect may fail to reach statistical significance if the data are limited.

3. CASE STUDY 17 – SOFTWARE APPLICATION RESPONSE TIME

Problem Statement

A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004. At $\alpha = 0.01$, determine whether there is sufficient evidence to support the company's claim.

Given Data

Item	Value
Company's claim	Average response time < 2 seconds
Reference value	2 seconds
p-value	0.004
α	0.01
Test direction	Left-tailed

Step 1 – Hypotheses

$$H_0: \mu \geq 2 \text{ seconds}$$

$$H_1: \mu < 2 \text{ seconds}$$

Here μ is the true population mean response time. The alternative hypothesis directly represents the company's claim.

Step 2 – Compare p-value with α

$$0.004 < 0.01$$

The p-value is smaller than the significance level.

Step 3 – Decision

Reject H_0 . The result is statistically significant at the 1% significance level.

Final Answer

There is sufficient statistical evidence at $\alpha = 0.01$ to support the company's claim that the application has an average response time of less than 2 seconds.

4. CASE STUDY 17 – DETAILED STATISTICAL

INTERPRETATION

The p-value of 0.004 provides strong evidence against the null hypothesis at the 1% significance level. Because 0.004 is smaller than 0.01, the observed evidence is sufficiently inconsistent with the baseline condition to reject H_0 .

The test is directional because the company specifically claims that the mean is below two seconds. Therefore, the alternative hypothesis is $\mu < 2$.

The conclusion concerns the population mean, not every individual response. Some requests may still take longer than two seconds because of workload, network conditions, hardware, database operations, or other factors.

Data Science Interpretation

For a production system, the hypothesis test can be complemented by median response time, standard deviation, percentile response times, and the percentage of requests exceeding the service target. These measures provide a more complete performance profile.

Overall, the statistical evidence supports the company's stated average-response-time claim at the 1% significance level.

5. CASE STUDY 18 – DIGITAL ADVERTISEMENT CLICK-THROUGH RATE

Problem Statement

A digital marketing team launches a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the rate was 6%. After the campaign, a hypothesis test produces a p-value of 0.12. What should the data science team conclude at the 5% significance level?

Item	Value
Baseline CTR	6%
p-value	0.12
α	0.05
Question	Does the new advertisement increase CTR?

Hypotheses

$$H_0: p \leq 0.06$$

$$H_1: p > 0.06$$

Decision Rule

$$0.12 > 0.05$$

The p-value is greater than α , so the null hypothesis is not rejected.

Final Answer

At the 5% significance level, there is insufficient statistical evidence to conclude that the new advertisement increases the click-through rate above 6%.

6. CASE STUDY 18 – DETAILED STATISTICAL

INTERPRETATION

The p-value of 0.12 does not meet the 5% threshold. Therefore, the observed evidence is not strong enough to establish a statistically significant increase in click-through rate.

This does not prove that the advertisement has no effect. It means that the available data do not provide sufficient evidence for an increase at $\alpha = 0.05$.

Business Recommendation

The marketing team should not declare the new advertisement a statistically validated success based on this test alone. Additional observations, a longer controlled experiment, or further A/B testing could provide more evidence.

The team should consider sample size, conversion counts, confidence intervals, effect size, experiment duration, and possible differences between customer segments.

Plain-Language Conclusion

The evidence is inconclusive at the 5% significance level. The team should not claim a statistically significant improvement in CTR from this test.

7. CASE STUDY 19 – MACHINE-LEARNING CLASSIFICATION

ACCURACY

Problem Statement

A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01. Explain what the p-value means and whether the team should conclude that the new model performs significantly better.

Item	Value
Old accuracy	82%
New accuracy	85%
Observed improvement	3 percentage points
p-value	0.01
α	Not explicitly stated

Meaning of $p = 0.01$

Assuming the null hypothesis of no true improvement is correct, a p-value of 0.01 indicates a 1% probability of obtaining evidence at least as extreme as the observed result, under the test assumptions.

Hypotheses

H_0 : The new model is not better than the old model

H_1 : The new model performs better than the old model

Decision

Because the question does not specify α , the decision must be stated relative to the chosen significance level. At $\alpha = 0.05$, $0.01 < 0.05$, so the result is statistically significant and H_0 is rejected.

At $\alpha = 0.01$, $p = \alpha$, so the usual rule $p \leq \alpha$ also leads to rejection.

Final Answer

At the commonly used 5% significance level, the evidence supports the conclusion that the new model performs significantly better.

8. CASE STUDY 19 – DETAILED STATISTICAL

INTERPRETATION

The observed accuracy increases from 82% to 85%, a three-percentage-point improvement. The statistical test evaluates whether such an observed difference is sufficiently unlikely under the null hypothesis.

At $\alpha = 0.05$, the p-value of 0.01 is below the threshold. Therefore, the improvement is statistically significant.

However, statistical significance is different from practical significance. The team should consider whether a three-point improvement justifies additional computational cost, latency, engineering effort, or operational complexity.

Recommended Model Evaluation

Before deployment, the team should use an appropriate held-out test set or validation design and examine other relevant metrics such as precision, recall, F1-score, confusion matrices, calibration, robustness, and generalization.

For high-impact systems, statistical significance should be only one part of the deployment decision.

9. CASE STUDY 20 – FACTORY DEFECT RATE

Problem Statement

A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01. Interpret both results and explain whether the company's claim should be rejected.

Item	Value
Claimed defect rate	2%
95% CI	(2.4%, 3.6%)
p-value	0.01
Confidence level	95%
Corresponding α	0.05

Confidence Interval Interpretation

The 95% confidence interval extends from 2.4% to 3.6%. The claimed value of 2% is below the lower endpoint, so 2% is not contained in the reported interval.

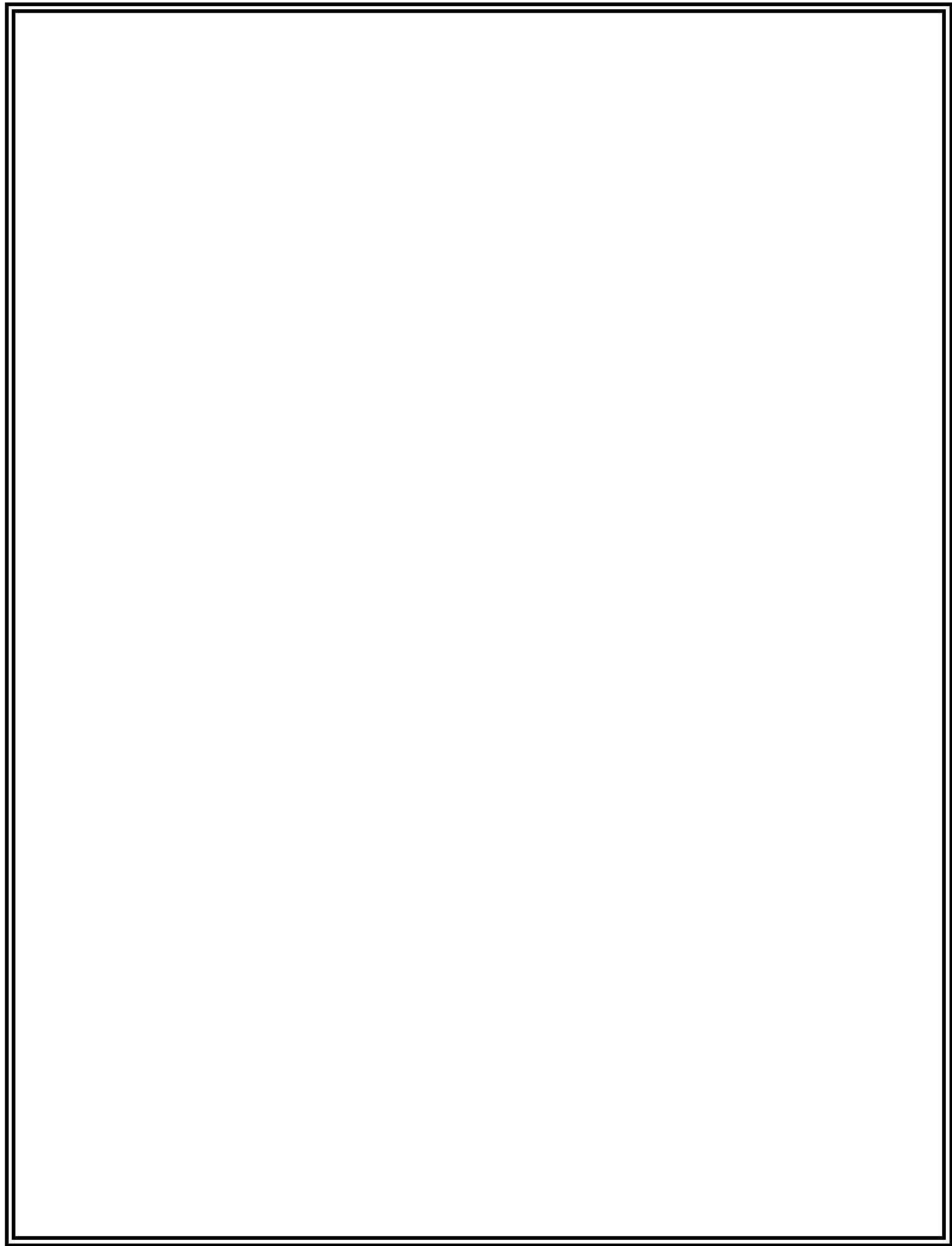
Hypothesis Test Interpretation

$$0.01 < 0.05$$

The p-value is below 0.05, so the null hypothesis representing the 2% claim is rejected at the 5% significance level.

Overall Decision

Both the confidence interval and p-value point in the same direction. The company's claim of a 2% defect rate should be rejected at the 5% significance level.



10. CASE STUDY 20 – CI + P-VALUE ANALYSIS

The confidence interval describes a range of plausible population defect rates, while the p-value provides a formal test against the claimed value.

Because the entire interval from 2.4% to 3.6% lies above 2%, the data indicate that the true defect rate is likely higher than the claimed 2%. The p-value of 0.01 independently provides statistically significant evidence against the claim.

Operational Interpretation

If the internal target is 2%, an estimated defect rate in the 2.4%–3.6% range is a quality-control concern. The organization should investigate manufacturing processes, supplier quality, inspection procedures, equipment calibration, and sources of defective output.

Conclusion

The company's 2% claim should be rejected at the 5% significance level. Both statistical approaches provide consistent evidence that the defect rate is higher.

11. CASE STUDY 21 – ONLINE SHOPPING CUSTOMER SPENDING

Problem Statement

An online shopping company wants to determine whether a new user-interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002. Interpret the confidence interval and p-value, and provide a business recommendation.

Item	Value
95% CI for difference	(₹120, ₹480)
p-value	0.002
Confidence level	95%
α	0.05
Question	Does the new UI increase average spending?

Confidence Interval Interpretation

The 95% confidence interval ranges from ₹120 to ₹480. The entire interval is positive, so the estimated difference favors the new interface.

Under the assumptions of the statistical procedure, the interval indicates a positive population difference, with the reported range corresponding to an increase of ₹120 to ₹480 per customer.

P-value Interpretation

$$0.002 < 0.05$$

The p-value is substantially below 0.05. Therefore, the result is statistically significant and provides evidence that the new user-interface design increases average spending.

Business Recommendation

The company should strongly consider adopting the new interface, subject to practical validation. The statistical evidence supports a positive effect.

Before full rollout, the company should check whether the improvement remains stable across customer segments, devices, markets, and time periods. It should also monitor returns, cancellations, complaints, customer satisfaction, and implementation cost.

12. CASE STUDY 21 – DETAILED BUSINESS INTERPRETATION

The confidence interval and p-value provide complementary evidence. The confidence interval is entirely above zero, indicating a positive effect across the reported range. The p-value of 0.002 indicates strong evidence against a no-increase null hypothesis.

Business Meaning of the Interval

The interval from ₹120 to ₹480 gives decision makers information about the possible magnitude of the increase. It does not mean that every customer will spend ₹120 to ₹480 more; it concerns the population-level difference estimated by the statistical procedure.

Recommended Action

A controlled rollout or continued A/B testing is appropriate. The company can monitor average order value, revenue per visitor, conversion rate, retention, refund rate, and satisfaction.

The company should compare expected incremental revenue with the development, testing, maintenance, and operational costs of implementing the new interface.

Conclusion

At the 5% significance level, the p-value supports a statistically significant increase, while the 95% confidence interval supports a positive difference of ₹120 to ₹480. The evidence favors adoption after practical validation.

13. COMPARISON OF STATISTICAL RESULTS

The supplied problems demonstrate two major uses of statistical inference: hypothesis testing using p-values and combined interpretation of confidence intervals with p-values.

Case	Result	Decision / Interpretation
17 – Response Time	$p = 0.004, \alpha = 0.01$	Reject H_0 ; sufficient evidence supports mean response time < 2 seconds.
18 – Advertisement CTR	$p = 0.12, \alpha = 0.05$	Fail to reject H_0 ; insufficient evidence of an increase above 6%.
19 – ML Accuracy	$p = 0.01$	At $\alpha = 0.05$, statistically significant improvement from 82% to 85%.
20 – Defect Rate	95% CI = 2.4%–3.6%; $p = 0.01$	Reject 2% claim; both methods indicate a higher defect rate.
21 – Customer Spending	95% CI = ₹120–₹480; $p = 0.002$	Positive, statistically significant increase; consider UI adoption.

Key Observations

Cases 17, 20, and 21 provide statistically significant evidence in favor of the stated effect or against the baseline claim. Case 18 does not provide sufficient evidence at the 5% level. Case 19 is significant at common levels such as 5%, but the question does not explicitly specify α , so the conclusion should state the selected significance level.

P-value and Confidence Interval

A p-value primarily answers a testing question: is the evidence sufficiently inconsistent with the null hypothesis? A confidence interval adds information about the direction and plausible magnitude of the effect.

14. PRACTICAL DATA SCIENCE IMPLICATIONS

Hypothesis testing is relevant to data science because real-world systems require decisions under uncertainty. A software engineer may need to determine whether an optimization reduces response time. A marketing analyst may need to determine whether a campaign improves engagement. A machine-learning team may need to determine whether a new model is genuinely better than a baseline.

1. Software Engineering

For response-time analysis, statistical testing can help distinguish genuine improvements from random variation. It should be complemented by percentile monitoring because an average can hide slow requests.

2. Digital Marketing

For advertising experiments, p-values can help determine whether an observed CTR increase is statistically supported. Effect size and business value should also be evaluated.

3. Machine Learning

Model comparisons require appropriate evaluation designs. Accuracy differences may depend on the dataset and may be insufficient when classes are imbalanced. Additional metrics are often necessary.

4. Manufacturing

Confidence intervals for defect rates can support quality-control decisions. If a target lies outside a confidence interval, the organization has evidence that the process may not meet the target.

5. E-commerce

For interface experiments, confidence intervals provide information about the possible magnitude of changes in customer spending and can support business planning.

Statistical Significance vs Practical Significance

A small p-value does not automatically mean an effect is important to the business. Large samples can produce small p-values for small effects. Decisions should therefore consider effect size, confidence intervals, cost, risk, and operational consequences.

Importance of Data Quality

All conclusions depend on data quality and the assumptions behind the test. Appropriate sampling, experimental design, independence where required, correct test selection, and avoidance of systematic bias are essential.

15. CONCLUSION AND QUICK REFERENCE

This report analyzed the supplied case studies involving p-values and confidence intervals. The cases demonstrate how statistical inference can be applied to software performance, advertising, machine-learning model evaluation, manufacturing quality, and online shopping.

Case 17: $p = 0.004$ is below $\alpha = 0.01$. Reject H_0 and support the claim that the mean response time is below two seconds.

Case 18: $p = 0.12$ is above $\alpha = 0.05$. Fail to reject H_0 ; there is insufficient evidence that the advertisement increased CTR above 6%.

Case 19: $p = 0.01$. At $\alpha = 0.05$, reject H_0 and conclude that the new model is statistically significantly better.

Case 20: The 95% CI (2.4%, 3.6%) excludes 2%, and $p = 0.01$. Reject the 2% defect-rate claim.

Case 21: The 95% CI (₹120, ₹480) is entirely positive and $p = 0.002$. The evidence supports a statistically significant increase in average spending.

Quick Reference

$p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$

$p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$

A p-value is not the probability that H_0 is true. Failing to reject H_0 does not prove H_0 . Statistical significance does not automatically imply practical significance. Confidence intervals provide information about plausible effect values and are especially useful for understanding the direction and magnitude of an effect.

Overall, the strongest data science decisions combine statistical evidence with sound experimental design, effect-size interpretation, practical importance, and domain knowledge.